

GSPS Implementation Blueprint

Gann-Centered Scanner System With Sara Sniper Strategy, Vortex/Digital-Root Coordinates, and Evidence-Gated Signals

Version: 1.0

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Primary product name: Guided Stock Precision Scanner (GSPS)

Former name: Gann Sniper Protocol Scanner

Audience: Claude implementation agent, engineers, product architects, quant researchers, and QA reviewers

1. Mission and governing doctrine

Build GSPS as a market-scanning and trade-planning system whose organizing principle is the practical, documented body of W. D. Gann's market methodology.

1.1 The Gann Sun model

Gann is the **sun** of GSPS. All other modules orbit it. The system must not become a random indicator bundle and must not reduce Gann to a decorative "3/6/9" label.

GSPS must operationalize the parts of Gann's work that are observable, reproducible, and testable:

1. Price structure and trend.
2. Supply and demand expressed through price and volume.
3. Accumulation and distribution over time.
4. Market position relative to prior highs, lows, ranges, and resistance/support.
5. Time elapsed within a swing, range, or trend.
6. Instrument-specific behavior ("habits of stocks").
7. Confirmation before entry.
8. Patience and no-trade conditions when evidence is unclear.
9. Defined stop loss, risk limitation, and post-trade review.

Gann's "vibration," harmony, natural law, periodicity, and cyclic language must be represented as **researchable hypotheses and market-state features**, not as guaranteed causal laws.

1.2 Supporting orbiting modules

The following modules support the Gann core:

- Sara Sniper Strategy: named GSPS execution strategy; its precise existing rules must be preserved, inventoried, and integrated into the confirmation/execution layer.
- Digital Root (DR): mandatory whenever Gann is mentioned in GSPS outputs.
- Vortex map: 1-2-4-8-7-5 flow loop, 3-6 polarity relationship, 9 completion node, and 1 renewal node.
- Conventional technical features only where measurable and validated: ATR, relative volume, anchored VWAP, moving-average trend state, prior swing levels, price range, relative strength, breadth/sector confirmation, volatility percentile, and optional volume-at-price.
- Risk engine: hard stops, position sizing, loss guardrails, trade eligibility, and tier-specific permissions.

1.3 Product promise

GSPS is a transparent market-state scanner and decision-support system. It identifies:

- Confirmed pullbacks within trends.
- Potential mean reversions after rejected or climactic moves.
- Momentum continuations after accepted, high-participation moves.
- Sideways/range opportunities where rule conditions are met.
- No-trade states where the market lacks sufficient evidence.

GSPS must never claim that a number, Gann coordinate, Vortex state, or indicator guarantees an outcome.

2. Non-negotiable terminology

2.1 Digital-root convention: active values only

GSPS operates with digital roots 1 through 9 only.

- 0 means absence, non-manifestation, missing, invalid, or unusable input.
- A valid positive number divisible by 9 has active root 9, not root 0.
- Never display 0 as a Gann node, harmonic node, signal root, or market state.
- Never convert missing/null/invalid data to root 9.

Mathematical formula for a positive integer n :

```
def digital_root_1_to_9(n: int) -> int:  
    if n <= 0:
```

```
        raise ValueError("GSPS digital roots require a positive integer")
    return 1 + ((n - 1) % 9)
```

Compatibility version for a valid numeric value that must be decimal-stripped per the legacy Gann Protocol:

```
def calculate_gann_dr(value) -> int | None:
    clean = "".join(ch for ch in str(value) if ch.isdigit())
    if not clean:
        return None
    integer_value = int(clean)
    if integer_value == 0:
        return None
    return digital_root_1_to_9(integer_value)
```

2.2 Data integrity rule

Do not use raw formatted display prices as the primary predictive input. A root derived from \$123.45 can change if quote precision changes, even without meaningful economics.

Use normalized integer measures. Preserve the legacy decimal-strip method as a separately labeled `display_price_dr` feature if desired, but do not let it alone determine a trade.

Every DR field must include:

```
raw_value
normalization_method
integer_value
mod9_residue
active_digital_root
input_timestamp
source_timeframe
feature_version
```

2.3 Gann labels

Use this consistent language in UI, reports, API, and logs:

```
Gann Digital Root: 1-9 only
Root 9: Completion / Culmination Node
Root 1: Renewal / Initiation Node
Complementary Root: Root that resolves with another active root to 9
Polarity Pair: One of 1-8, 2-7, 3-6, 4-5, or 9-9
Gann Coordinate: A transparent calculated price/time/volume state, not a forecast guess
Experimental: Feature not yet promoted by out-of-sample validation
Validated: Feature that passed the configured promotion criteria
No Trade: Valid output state, not an error
```

3. What Gann documented: requirements translation

The attached source material must be treated as the project’s historical foundation.

3.1 Direct Gann-derived operating requirements

Gann principle	GSPS requirement
Price movements depend on supply and demand	Require price and volume/participation data in every trade-eligible signal
Major moves require time for accumulation or distribution	Measure time in range, bars since pivot, and range/volume structure
Volume needed to move an issue depends on shares/float	Include liquidity, dollar volume, float/market-cap context when available
Trend must be distinguished from minor reactions	Build multi-timeframe trend and swing structure engine
News is often discounted, while surprise matters	Add event-aware state and require price reaction confirmation rather than narrative-only signals
Traders are harmed by hope, fear, and overtrading	Include no-trade output, cooldowns, fixed invalidation, and coaching explanations
Stops must be placed when the trade is made	No live/paper trade plan can exist without a deterministic stop/invalidation
Focus on selected instruments and their habits	Build instrument behavior profiles and watchlist specialization
Be out of the market when no trend is clear	Neutral/no-trade is mandatory; never fabricate directional certainty

3.2 Historical boundary

Do not claim that Gann disclosed the GSPS decimal-strip modulo-9 formula, a mandatory 3/6/9 target formula, or a universal fixed “vibration” calculation. The historical source uses language of cycles, vibration, individual stock characteristics, support/resistance, and time but does not disclose a complete reproducible calculation.

Implement Gann-related numeric elements as explicit GSPS hypotheses with formulas, versioning, and tests.

4. The GSPS winning formula

4.1 Formula statement

A GSPS trade candidate becomes actionable only when a Gann-centered market state has measurable confluence:

```
GSPS Actionable Setup
= Market Regime
+ Higher-Timeframe Trend
+ Price Structure and Location
+ Supply/Demand Confirmation
```

- + Time-in-Structure Context
- + Volatility Regime
- + Gann/Vortex/DR Coordinate Confluence
- + Sara Sniper Execution Trigger
- + Deterministic Risk Plan
- + Liquidity and Event Filters

Every factor must be present in the signal record. Missing factors lower confidence or produce NO_TRADE.

4.2 Mandatory condition hierarchy

The hierarchy is intentionally ordered. Lower-priority elements may not override higher-priority invalidation.

1. Data integrity and tradability.
2. Risk gate and user eligibility.
3. Market regime and higher-timeframe trend.
4. Price structure and location.
5. Supply/demand, volume, and volatility behavior.
6. Time and cycle context.
7. Gann/Vortex/DR coordinate context.
8. Sara Sniper trigger and entry mechanics.
9. Position management, targets, stop, and expiry.

Example: A digital-root completion node cannot override a bearish breakdown with increasing downside volume. It may create a reversion alert, but no long entry is allowed without the defined reclaim/confirmation trigger.

4.3 Strategy families

GSPS must classify signals into one mutually exclusive primary type:

- MOMENTUM_CONTINUATION
- MEAN_REVERSION
- PULLBACK_CONTINUATION
- RANGE_ROTATION
- NO_TRADE

A signal can list secondary tags, but must not simultaneously report contradictory primary strategies.

5. System architecture

5.1 Module map

```
Market Data Layer
-> Data Normalization Layer
-> Pivot / Trend / Structure Engine
-> Supply-Demand / Volume / Liquidity Engine
-> Time and Cycle Engine
-> Gann Coordinate Engine
-> Vortex and Digital Root Engine
-> Indicator Ensemble
-> Sara Sniper Strategy Engine
-> Strategy Classifier
-> Risk and Position-Sizing Engine
-> Signal Ranking / Tier Entitlements
-> Explainability and Audit Ledger
-> Backtest / Walk-Forward Validation Platform
```

5.2 Data model

Create normalized tables or equivalent domain entities:

```
instrument
bar
corporate_action
instrument_profile
pivot
trend_state
volume_state
volatility_state
market_regime
gann_coordinate
vortex_state
digital_root_feature
strategy_signal
trade_plan
risk_plan
signal_outcome
backtest_run
feature_registry
experiment_registry
model_version
audit_event
```

5.3 Required audit fields

Every generated signal must be reproducible from stored data. Persist:

```
signal_id
instrument_id
```

```

timeframe
bar_close_timestamp
data_vendor
all OHLCV inputs used
market regime inputs
selected pivots and pivot confirmation timestamps
feature values and formulas
parameter-set version
strategy version
entry rule
entry price or trigger
stop price and invalidation reason
target rules and levels
time stop
estimated fees/spread/slippage
liquidity score
risk amount and size rule
user tier / paper-live mode
why_actionable[]
why_not_actionable[]

```

6. Data normalization

6.1 Canonical price measures

Implement these as positive integer measures where possible:

```

price_displacement_ticks = round(abs(last_price - anchor_price) / tick_size)
time_displacement_bars = max(1, current_bar_index - anchor_bar_index)
bar_range_ticks = max(1, round((high - low) / tick_size))
atr_ticks = max(1, round(atr_n / tick_size))
level_distance_ticks = max(1, round(abs(last_price - key_level) / tick_size))
relative_volume_index = max(1, round(100 * volume / avg_volume_n))
dollar_volume_index = max(1, round(dollar_volume / dollar_volume_unit))

```

For negative directional measures, store direction separately. Do not compute a digital root over a negative string.

```

direction: UP / DOWN / FLAT
magnitude_integer: positive normalized integer
active_digital_root: 1-9

```

6.2 Asset-class adapters

Asset class	Primary normalization
Equities / ETFs	Tick size, adjusted OHLCV, dollar volume, float/market-cap profile
Futures	Tick value, contract multiplier, roll-adjusted or contract-specific history, session calendar

Asset class	Primary normalization
Forex	Pip/pipette scale, session boundaries, spread/rollover assumptions
Crypto	Exchange-specific tick size, 24/7 session model, spot/perpetual distinction, funding rates
Options	Underlying-state signal plus option strike/expiry/Greeks/liquidity adapter; never root-only option selection

7. Vortex and digital-root engine

7.1 Mathematical state mapping

Implement the active 1-9 root convention.

```
VORTEX_FLOW = [1, 2, 4, 8, 7, 5]
POLARITY_PAIRS = {
    1: 8,
    8: 1,
    2: 7,
    7: 2,
    3: 6,
    6: 3,
    4: 5,
    5: 4,
    9: 9,
}

def gann_complement(root: int) -> int:
    if root not in range(1, 10):
        raise ValueError("root must be 1 through 9")
    return POLARITY_PAIRS[root]

def resolves_to_completion(root_a: int, root_b: int) -> bool:
    return digital_root_1_to_9(root_a + root_b) == 9

def vortex_class(root: int) -> str:
    if root in VORTEX_FLOW:
        return "VORTEX_FLOW"
    if root in (3, 6):
        return "POLARITY_AXIS"
    if root == 9:
        return "COMPLETION_NODE"
    raise ValueError("root must be 1 through 9")
```

7.2 Required root features

Calculate and persist at minimum:

```
price_displacement_dr
time_displacement_dr
swing_range_dr
bar_range_dr
```



```
atr_dr
relative_volume_dr
level_distance_dr
display_price_dr (legacy/context-only)
```

7.3 Confluence types

```
NO_CONFLUENCE
COMPLEMENTARY_PAIR
COMPLETION_PAIR
THREE_SIX_POLARITY
NINE_COMPLETION
VORTEX_FLOW_TRANSITION
ONE_RENEWAL_TRANSITION
MULTI_FACTOR_CONFLUENCE
```

7.4 Safety rule

DR/Vortex features are initially EXPERIMENTAL. They may:

- Add tags.
- Rank research candidates.
- Highlight candidate price/time windows.
- Contribute to a confidence score.

They may not by themselves:

- Create a live-trade entry.
- Override a stop.
- Override trend or risk gates.
- Create exact price targets without a separate deterministic price-level formula.

8. Gann coordinate engine

8.1 Purpose

The Gann Coordinate Engine must turn price, time, volume, range, and DR/Vortex state into transparent, inspectable candidate zones.

8.2 Anchor selection

Use objective, configurable pivot rules. Initial implementation:

```
Pivot low:
A low that is lower than N bars before and N bars after.
It becomes usable only after the N future confirmation bars close.
```

Pivot high:

A high that is higher than N bars before and N bars after.

It becomes usable only after the N future confirmation bars close.

Persist both:

```
pivot_occurrence_timestamp  
pivot_confirmation_timestamp
```

Never allow the system to trade using a pivot before it was confirmed.

8.3 Candidate price coordinates

Generate and label candidate levels from:

- Confirmed swing high and low.
- Prior daily/weekly/monthly high and low.
- Range midpoint and range fractions.
- Measured move from confirmed swing range.
- Anchored VWAP from major pivot/event.
- Optional Square-of-Nine module only after formula and testing specification are approved.
- Volatility-based bands using ATR.

Every coordinate must store:

```
coordinate_type  
anchor_id  
formula_description  
parameter_values  
price_level  
side: SUPPORT / RESISTANCE / NEUTRAL  
confidence_basis  
research_status
```

8.4 Time coordinates

Initial time coordinates must be simple and testable:

- Bars since confirmed pivot.
- Trading days since confirmed pivot.
- Time spent in a range.
- Duration of current trend leg.
- Time symmetry ratio: current swing duration / prior swing duration.
- Predefined event windows.

Do not claim universal cycle length. Store candidate time windows as alerts and test them by asset, timeframe, and regime.

8.5 Gann angle design

Do not use chart-screen pixels. Define normalized slopes such as:

```
price change per bar in ATR units  
percent price change per bar  
log-price change per bar
```

A 1x1-type reference needs a declared unit. Initial normalized version:

```
normalized_slope = (price_t - anchor_price) / (atr_at_anchor * bars_since_anchor)
```

Only expose Gann-angle labels when the scale and anchor are explicitly stored.

9. Supply-demand, volume, and volatility engine

9.1 Required calculations

```
relative_volume = current_volume / rolling_average_volume  
relative_dollar_volume  
ATR  
ATR percentile  
true-range percentile  
rolling realized volatility  
close_location_value  
up-volume/down-volume proxies  
volume persistence: elevated volume count in recent bars  
range compression/expansion state  
liquidity score  
spread estimate or observed spread where available
```

9.2 State classification

Implement these states:

```
QUIET_ACCUMULATION  
QUIET_DISTRIBUTION  
ACCEPTED_UPSIDE_EXPANSION  
ACCEPTED_DOWNSIDE_EXPANSION  
REJECTED_UPSIDE_EXPANSION  
REJECTED_DOWNSIDE_EXPANSION  
LOW_PARTICIPATION_RANGE  
UNCLASSIFIED
```

9.3 Initial deterministic definitions

Use configuration rather than hard-coded constants. Starting defaults may be:

```
relative volume elevated: >= 1.5x 20-bar average
relative volume climactic: >= 2.0x 20-bar average
range expansion: true range >= 75th percentile of trailing 60 bars
range climax: true range >= 90th percentile of trailing 60 bars
bullish acceptance: close location >= 0.75
bearish acceptance: close location <= 0.25
```

Do not promote these defaults as universal truths. Backtest them by instrument and timeframe.

10. Strategy engines

10.1 Mean-reversion engine

Purpose: identify exhausted or rejected moves, not merely “oversold” or “overbought” conditions.

Long mean-reversion candidate

1. Price is at/near a mapped support coordinate or materially extended below a defined coordinate.
2. Downside volatility is expanded or climactic.
3. Selling volume is elevated/climactic or shows exhaustion/rejection pattern.
4. Price structure shows failed breakdown, reclaim, reversal, or higher low.
5. Gann/DR/Vortex completion or polarity condition exists as an experimental confluence.
6. Sara Sniper long trigger confirms.
7. Stop/invalidation is defined below the exhaustion low or invalidation coordinate.
8. Liquidity/event/risk gates pass.

Short mean-reversion candidate

Mirror the long logic around mapped resistance, upside extension, climactic buying, failed breakout, rejection, and a bearish Sara Sniper trigger.

10.2 Momentum-continuation engine

Purpose: identify accepted, persistent moves with sustained participation; do not confuse intensity with exhaustion.

Long continuation candidate

1. Higher timeframe trend is bullish.
2. Price breaks or reclaims a defined resistance/support coordinate.
3. Relative volume is elevated and persists.
4. Range expands while bars close near their highs.
5. Price remains accepted above breakout level or passes retest.

6. Gann/DR/Vortex state supports ranked confluence but does not independently cause a signal.
7. Sara Sniper continuation trigger confirms.
8. Stop lies below retest low / higher low / volatility invalidation.

Short continuation candidate

Mirror the logic for bearish trend, breakdown acceptance, lower-range closes, failed reclaim, and bearish trigger.

10.3 Pullback-continuation engine

Purpose: identify controlled pullbacks within an established trend.

1. Higher-timeframe trend is established.
2. Pullback returns to a mapped level, trend reference, or anchored VWAP zone.
3. Pullback volume contracts relative to impulse volume or rejection occurs at the start of the pullback.
4. Trend resumes through a defined trigger.
5. DR/Vortex/time coordinate is logged and scored.
6. Stop is structural and deterministic.

10.4 Range-rotation engine

Purpose: identify rule-compliant trades between established range support and resistance.

1. Market regime is range/neutral, not strong directional trend.
2. Defined support and resistance exist with adequate historical touches.
3. Entry occurs only on confirmation at a boundary.
4. Stop is beyond range invalidation.
5. Target is range midpoint or opposing boundary depending on risk plan.
6. Avoid entries during abnormal event volatility or confirmed breakout acceptance.

10.5 No-trade engine

Return NO_TRADE when any of these applies:

- No confirmed trend/structure.
- Insufficient liquidity.
- Excessively wide spread.
- Earnings/event risk conflicts with user rules.
- No deterministic stop.
- Signal conflicts across required layers.
- Risk/reward or expected value gate fails.
- User tier cannot trade the strategy/timeframe.
- Data missing, stale, or invalid.
- Numeric confluence exists without price/volume confirmation.

11. Sara Sniper Strategy integration

The Sara Sniper Strategy is a named GSPS strategy and must be an independently versioned module.

11.1 Immediate engineering task

Search the repository, prompt records, strategy documents, database, and prior implementation for all rules associated with:

```
Sara Sniper Strat
Sara Sniper Strategy
Sara Sniper
```

Create:

```
/docs/strategy-registry/sara-sniper.md
/src/strategies/sara_sniper/
/tests/strategies/test_sara_sniper_*.py
```

Do not invent rules if they are not present. If exact legacy rules cannot be found, create a SARASniperSpecRequired state and list the unresolved fields.

11.2 Required strategy interface

```
class StrategyResult:
    strategy_id: str
    strategy_version: str
    signal_type: str
    direction: str
    status: str # WATCH / DEVELOPING / ACTIONABLE / NO_TRADE
    entry_trigger: float | None
    stop_loss: float | None
    targets: list[float]
    time_stopBars: int | None
    confidence_score: float
    conditions_met: list[str]
    conditions_failed: list[str]
    feature_snapshot_id: str
```

Sara Sniper should consume the Gann core state rather than bypass it:

```
Gann coordinates define context and attention zones.
Volume/volatility classify acceptance or rejection.
Sara Sniper defines the exact tactical confirmation and trade-plan construction.
Risk engine approves or rejects execution.
```

12. Indicator ensemble policy

Indicators are supporting instruments, not the sun.

12.1 Approved initial candidates

Indicator / measure	Purpose
ATR	Volatility normalization, stops, extension, position sizing
Relative volume	Participation and supply/demand confirmation
Anchored VWAP	Acceptance/rejection and institutional reference zone
Moving averages	Trend/regime context, not stand-alone entries
Prior swing levels	Support/resistance and structural invalidation
Price range / true range	Expansion, compression, climax detection
Relative strength vs SPY/sector	Context and leadership/lagging confirmation
Volume-at-price	Optional acceptance/rejection zones when data available
RSI	Optional secondary exhaustion condition only; never sole signal
MACD	Optional trend/momentum corroboration only; never sole signal

12.2 Indicators excluded as automatic authority

Do not allow any one indicator, oscillator, number root, or visual chart angle to independently generate a live signal.

All new indicators require:

```
feature specification
reason for inclusion
baseline comparison
out-of-sample evaluation
cost-adjusted performance test
ablation result
promotion/rejection decision
```

13. Risk, targets, and trade management

13.1 Non-negotiable risk rules

```
No trade plan without entry, stop, invalidation explanation, target rule, and time s
Stop must be created when the plan is created.
Root/Vortex state cannot widen or remove a stop.
Signal is invalid if the structural premise fails.
```

13.2 Position sizing

```
risk_per_share = abs(entry_price - stop_price)
max_dollar_risk = account_equity * user_risk_fraction
position_size = floor(max_dollar_risk / (risk_per_share + estimated_slippage_per_share))
```

13.3 Existing GSPS guardrails to preserve

- Novice users: swing trading only, up to three trades per trading day.
- Novice cooldown: activate when the user reaches three trades in a trading day or loses more than 18% of total funds within 48 hours.
- Loss notifications: notify when a single trade depletes 6%, 9%, or 15% of total funds.
- Hard warning: at 30% loss of total funds in a trade.
- Automatic close: at 50% loss of total funds in that trade, for live trading.
- Paper trading is exempt from live-execution restrictions.
- Stop-loss expansion/removal is Wall Street tier only, only after mandatory warning, and only after verified email and phone notification enrollment.
- Automation/live execution is restricted to Wall Street members within the dedicated Automation tab; paper trading is exempt.

13.4 Target philosophy

Targets must be derived from deterministic market structures, not only numeric labels.

Candidate target sources:

```
1R partial target
2R or marker-specific percentage target
Prior swing high/low
Mapped Gann price coordinate
Anchored VWAP / range midpoint
Measured move
Validated volatility projection
Trailing stop after target 1
```

Preserve the user's requirement for marker-specific percentage take-profit and stop-loss levels rather than a rigid universal 2:1 risk/reward rule. Target/stop percentages must be explicit per strategy marker and tested by asset class.

14. Signal scoring and tiers

14.1 Score by evidence, not mystique

Build a configurable score with contribution disclosure.

Example components:

```
trend_alignment_score
price_structure_score
level_quality_score
volume_confirmation_score
volatility_regime_score
time_context_score
gann_coordinate_score
digital_root_vortex_score
sara_sniper_trigger_score
liquidity_score
market_benchmark_score
event_risk_penalty
conflict_penalty
```

14.2 Score output

```
0-24: NO_TRADE
25-49: WATCH
50-69: DEVELOPING
70-84: ACTIONABLE
85-100: HIGH_CONFLUENCE
```

Thresholds are placeholders and must be calibrated through research. Store configuration by version.

14.3 Tier allocation

Honor existing scan output limits:

User tier	Maximum total setups per scan	Manual dashboard scans/day
Novice	6, ideally 3 long and 3 short	1
Pro	12	3
Expert	20	6
Wall Street	30	Unlimited

Rank primarily by eligibility, confidence, liquidity, risk quality, and diversification—not by numeric root alone.

15. Backtesting and research platform

15.1 Research question

Every DR/Vortex/Gann feature is a falsifiable hypothesis:

Does this feature improve a specified strategy's post-cost performance relative to the identical strategy without the feature?

15.2 Required baselines

For each strategy and asset/timeframe, compare:

Buy-and-hold or cash benchmark where appropriate.
Simple trend-following baseline.
Simple breakout baseline.
Simple mean-reversion baseline.
Same GSPS strategy with Gann/DR/Vortex feature removed.
Random-time matched control signals.
Prior-swing support/resistance strategy without DR/Vortex feature.

15.3 Required metrics

Trade count
Win rate
Average win
Average loss
Expectancy in R
Profit factor
Net return after costs
Maximum drawdown
Sharpe ratio
Sortino ratio
Calmar ratio
Turnover
Capacity/liquidity feasibility
Performance by regime
Performance by asset
Performance by timeframe
Stability under parameter perturbation

15.4 Bias controls

Must implement or document controls for:

Look-ahead bias
Survivorship bias
Delisting bias
Corporate-action bias

- Data-snooping bias
- Curve fitting
- Multiple testing
- Cherry-picking
- Selection bias
- Regime selection bias
- Unrealistic fill assumptions
- Ignored transaction costs
- Ignored market impact
- Future-known pivots

15.5 Validation requirements

- Development sample -> validation sample -> locked out-of-sample sample.
- Walk-forward test across multiple eras.
- Post-cost test.
- Block bootstrap or appropriate dependent-data resampling.
- Permutation tests for root labels/conditions.
- Confidence intervals.
- Multiple-testing correction when testing feature grids.
- Ablation: same model minus the candidate root/Vortex/Gann feature.
- Paper-trading reconciliation before live automation.

15.6 Promotion rule

A feature can move from EXPERIMENTAL to VALIDATED only if it:

- Improves a predeclared post-cost metric over a relevant baseline.
- Passes locked out-of-sample testing.
- Remains directionally stable across reasonable parameter changes.
- Does not depend on one instrument, one era, or one volatility regime.
- Does not materially worsen maximum drawdown/tail risk.
- Has adequate sample size.
- Has reproducible code and an audit trail.

Otherwise label it EXPERIMENTAL, INCONCLUSIVE, or REJECTED.

16. Asset-class and execution constraints

16.1 Equities and ETFs

Account for:

- Bid/ask spread
- Commissions/fees
- Slippage
- Market impact
- Short locate/borrow cost
- Corporate actions

- Delistings
- Earnings gaps
- Liquidity screens
- Point-in-time constituent membership

16.2 Futures

Account for:

- Contract specifications
- Tick value
- Contract multiplier
- Margin
- Roll construction and roll costs
- Session/calendar definition
- Limit moves
- Overnight liquidity

16.3 Forex

Account for:

- Pip convention
- Bid/ask spread
- Rollover/swap
- Session-specific liquidity
- Leverage
- Central-bank event exposure

16.4 Crypto

Account for:

- Exchange-specific liquidity
- 24/7 bar/session convention
- Maker/taker fees
- Funding rates
- Liquidation mechanics
- Perpetual vs spot distinction
- Custody/exchange risk

16.5 Options

Do not turn an underlying root condition directly into an option recommendation. Add:

- Underlying trade thesis
- Expiration selection rule
- Strike selection rule
- Delta/Gamma/Theta/Vega at entry

Implied-volatility state and event risk
Bid/ask spread and open interest
Maximum loss
Assignment/exercise handling
Theta decay and IV crush analysis
Comparison to underlying-only trade

17. User experience requirements

17.1 Required signal explanation

Each surfaced setup must answer in plain English:

What is price doing?
What is volume doing?
What is volatility doing?
What is the higher-timeframe trend?
Where is price relative to the mapped Gann coordinates?
What are the digital roots and Vortex classifications?
Is the root feature experimental or validated?
Why is this a reversion, continuation, pullback, range rotation, or no-trade state?
What exact event triggers entry?
What invalidates the thesis?
What are targets and their formulas?
What risks exist?

17.2 Required warning language

Use clear language:

“Digital-root and Vortex features are contextual classifications. They do not guarantee a trade outcome.”
“This setup is not trade-eligible until price confirms the trigger.”
“High volume can mean acceptance/continuation or exhaustion/reversal. The scanner displays both possibilities.”
“No Trade is an intentional protective outcome.”

17.3 Educational integration

The GSPS School tab must teach the user why a setup appears, using Mrs. Bear and Mr. Bull as instructional guides. Lessons must explain trend, support/resistance, volume, volatility, stop placement, reversion versus continuation, and experimental numeric features without promising certainty.

18. API contract example

```
{
  "ticker": "AAPL",
  "timeframe": "1h",
  "as_of": "2026-09-08T18:00:00-04:00",
  "primary_strategy": "MOMENTUM_CONTINUATION",
  "status": "ACTIONABLE",
  "direction": "LONG",
  "market_state": {
    "trend": "BULLISH",
    "structure": "ACCEPTED_UPSIDE_EXPANSION",
    "relative_volume": 1.82,
    "atr_percentile": 79,
    "liquidity_score": 96
  },
  "gann_context": {
    "anchor": {"type": "CONFIRMED_SWING_LOW", "price": 0.0, "timestamp": "..."},
    "price_displacement_ticks": 0,
    "price_dr": 8,
    "time_displacementBars": 0,
    "time_dr": 1,
    "relationship": "COMPLETION_PAIR",
    "vortex_class": "VORTEX_FLOW",
    "research_status": "EXPERIMENTAL"
  },
  "plan": {
    "entry_trigger": 0.0,
    "stop_loss": 0.0,
    "stop_reason": "Break below confirmed retest low",
    "targets": [0.0, 0.0],
    "time_stop_bars": 10,
    "position_size_rule": "Account risk / all-in cost-adjusted risk per share"
  },
  "explanation": {
    "conditions_met": [],
    "conditions_failed": [],
    "no_trade_conditions": [],
    "disclaimer": "Gann/DR/Vortex features are contextual and do not guarantee outcor
  },
  "audit": {
    "strategy_version": "...",
    "feature_version": "...",
    "parameter_set_id": "...",
    "data_snapshot_id": "..."
  }
}
```

Do not use zero prices in production payloads; they are placeholders in this example only.

19. Implementation milestones

Milestone 0: Discovery and inventory

- Locate the actual GSPS codebase, current branch, database schema, deployment configuration, and prior strategy docs.
- Search for all Gann, root, modulo, Vortex, harmonic, Sara Sniper, reversion, continuation, volume, volatility, stop, target, tier, and risk code.
- Produce a traceability matrix: existing / partial / absent / contradictory.
- Do not overwrite legacy logic until it is documented.

Milestone 1: Foundation services

- Implement canonical digital-root 1–9 service.
- Implement data normalization service.
- Implement feature registry and audit schema.
- Add unit tests for roots, complement pairs, zero/null handling, and precision invariance.

Milestone 2: Structure and market-state engines

- Build pivot confirmation.
- Build multi-timeframe trend state.
- Build support/resistance and anchored VWAP coordinates.
- Build relative volume, ATR, volatility percentile, liquidity, and acceptance/rejection states.

Milestone 3: Gann/Vortex coordinates

- Build price/time/range/volume DR features.
- Build complement/polarity/completion classification.
- Build transparent Gann coordinate ledger.
- Keep root-driven signals experimental.

Milestone 4: Strategy engines

- Implement mean reversion, momentum continuation, pullback continuation, range rotation, and no-trade logic.
- Integrate Sara Sniper once exact rules are inventoried.
- Build deterministic entry/stop/target/time-stop plans.

Milestone 5: Risk, entitlement, and UX

- Implement risk calculations and existing live/paper tier restrictions.
- Surface clear signal cards and no-trade explanations.
- Add educational explanations in GSPS School.

Milestone 6: Research platform

- Build event-driven backtester.
- Add realistic costs.
- Add benchmarks, ablation testing, walk-forward tests, and experiment registry.
- Produce validation reports for every promoted feature.

Milestone 7: Paper trading then controlled live use

- Reconcile signal records with simulated/paper fills.
- Evaluate realized slippage and adherence.
- Enable live execution only under existing Wall Street Automation tab controls.

20. Acceptance criteria

A release is not complete unless all of the following are true:

- Digital roots always produce 1–9 for valid positive inputs.
- Zero, null, missing, invalid, and stale values never become root 9.
- Every root is traceable to normalized input and formula.
- Every Gann-referenced output includes DR determination.
- No root/Vortex label can create a live entry by itself.
- Every actionable setup contains entry, stop, invalidation reason, targets, time stop, risk calculation, and explanation.
- No-trade output exists and is visible.
- Reversion and continuation classifiers are separately implemented and cannot be conflated.
- High volume/volatility is classified as accepted versus rejected using follow-through and price-location evidence.
- Sara Sniper rules are sourced and versioned, not guessed.
- Backtests prevent future-known pivots and include costs.
- Every experimental feature is visibly labeled.
- Every validated feature has a stored out-of-sample validation record.
- Existing GSPS tier and live-trading guardrails are enforced.

21. Final implementation instruction to Claude

Implement GSPS as a Gann-centered, Vortex-aware, digital-root-coordinate scanner whose purpose is to identify and explain high-quality market states—not to issue mystical certainty.

Use this decision law:

Gann establishes the operating philosophy:
price + time + volume + supply/demand + market position + discipline.

Vortex and DR establish the 1-9 coordinate language:
flow + polarity + completion + renewal.

Sara Sniper establishes tactical execution:
when a context becomes an exact entry, stop, target, and management plan.

Other indicators support, measure, and verify:
they never replace the Gann core.

Risk controls govern every trade:
no stop, no trade; no confirmation, no trade; no clarity, no trade.

Build the audit trail and test harness before promoting numerical or Gann-related claims into trade-eligible logic.